Cyber Security Internship - Interview Study Guide

1. What is an open port?

An **open port is a communication channel on a device that is actively listening for incoming network traffic** and allowed by the firewall to accept data.

2. How does Nmap perform a TCP SYN scan?

Nmap sends a **SYN packet** to the target host:

- **If Open:** The target replies with a **SYN-ACK**, and Nmap instantly drops the connection with a **RST packet** before it fully establishes.
- **If Closed:** The target replies directly with a **RST packet**.

3. What risks are associated with open ports?

- **Exploits:** Attackers can target unpatched vulnerabilities in the service listening behind the port.
- **Data Leakage:** Exposed ports reveal system identity and software versions to scanner programs.
- **Intrusion:** Weak or default credentials on open services can allow unauthorized access to the network.

4. Explain the difference between TCP and UDP scanning.

- **TCP Scanning:** Uses a connection-oriented handshake, providing fast, structural, and highly accurate port status responses.
- **UDP Scanning:** Uses connectionless packets and relies heavily on missing ICMP error messages, making it significantly slower and less reliable.

5. How can open ports be secured?

- **Disable Services:** Shut down or delete any application or background service that is not actively needed.
- **Apply Patches:** Keep the underlying software updated to resolve any discovered security flaws.
- **Restrict Access:** Configure firewalls to allow connections exclusively from trusted IP addresses.

6. What is a firewall's role regarding ports?

A firewall acts as a **traffic filter**. It monitors and blocks unauthorized data packets before they can interact with the network services running on your open ports.

7. What is a port scan and why do attackers perform it?

A port scan is a **network reconnaissance probe**. Attackers use it to discover active devices, locate open doors, and identify vulnerable targets within a network.

8. How does Wireshark complement port scanning?

Wireshark captures and dissects **raw packet traffic**. While Nmap tells you which ports are open, Wireshark lets you visually analyze the exact data packets exchanged during that scan.